

PROJECT REPORT:CREDIT CARD FRAUD DETECTION

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Abstract:

With the increasing use of online payments and digital banking services, credit card fraud has become a major concern for financial institutions and customers. Detecting fraudulent transactions manually is difficult because of the large number of transactions processed every day.

The aim of this project is to build a machine learning-based system that can identify fraudulent credit card transactions. The system uses transaction data to train an XGBoost model and applies SMOTE to handle the imbalance between genuine and fraudulent transactions. A Flask web application is developed to provide a simple interface where users can enter transaction details and obtain predictions instantly.

This project demonstrates how machine learning can be used to improve transaction security and assist in detecting suspicious activities efficiently.

Project Overview:

Credit card fraud detection is an important application of machine learning in the financial sector. Fraudulent transactions often occur in patterns that may not be easily recognized through manual monitoring. Machine learning algorithms can analyze historical transaction data and learn these patterns to make accurate predictions.

In this project, a fraud detection model is developed using transaction data. After training, the model is integrated with a web application so that users can enter transaction details and receive a prediction indicating whether the transaction is likely to be genuine or fraudulent.

Objectives:

- To develop a machine learning model for fraud detection.
- To analyze transaction data and identify fraud patterns.
- To improve detection accuracy using XGBoost.
- To handle class imbalance using SMOTE.
- To provide predictions through a web-based interface.
- To demonstrate a practical application of machine learning in finance.

Technologies Used:

Programming Language

Python

Libraries

Pandas

NumPy

Scikit-Learn

XGBoost

Imbalanced-Learn (SMOTE)

Joblib

Web Technologies

Flask

HTML

CSS

Methodology:

Data Collection

The credit card transaction dataset is collected and loaded into the system for analysis.

Data Preprocessing

The dataset is cleaned and prepared for model training. Necessary transformations and encoding techniques are applied to improve data quality.

Data Balancing

Since fraudulent transactions are much fewer than genuine transactions, SMOTE is used to balance the dataset and improve model learning.

Model Training

The XGBoost algorithm is trained using the processed dataset. The model learns patterns associated with fraudulent and legitimate transactions.

Prediction

The trained model predicts whether a transaction is fraudulent or genuine based on the input provided by the user.

Deployment

The model is integrated into a Flask application that provides real-time fraud prediction through a user-friendly interface.

Implementation

The project consists of two main modules.

Machine Learning Module

This module performs data preprocessing, model training, evaluation, and prediction generation. The trained model is saved and loaded whenever predictions are required.

Web Application Module

The Flask application acts as the user interface. Users enter transaction details through the web page, and the system displays the prediction result generated by the model.

Results and Screenshots:

The developed system successfully predicts whether a transaction is fraudulent or legitimate based on the provided transaction details.

Home Page



Credit Card Fraud Detection

AI-Powered Real-Time Fraud Detection using XGBoost & Flask

Electronics

☐  Foreign Transaction

☐  Location Mismatch

Detect Fraud

Reset

Legitimate Transaction:



LEGITIMATE TRANSACTION

Fraud Probability

0.01%

Risk Level

LOW

Recommendation

✓ Transaction approved. No suspicious activity detected.

Transaction Details

 Amount : ₹ 10000

 Transaction Hour : 10

 Merchant Category : Electronics

 Foreign Transaction : Yes

 Location Mismatch : No

 Device Trust Score : 100

 Transactions Last 24 Hours : 2

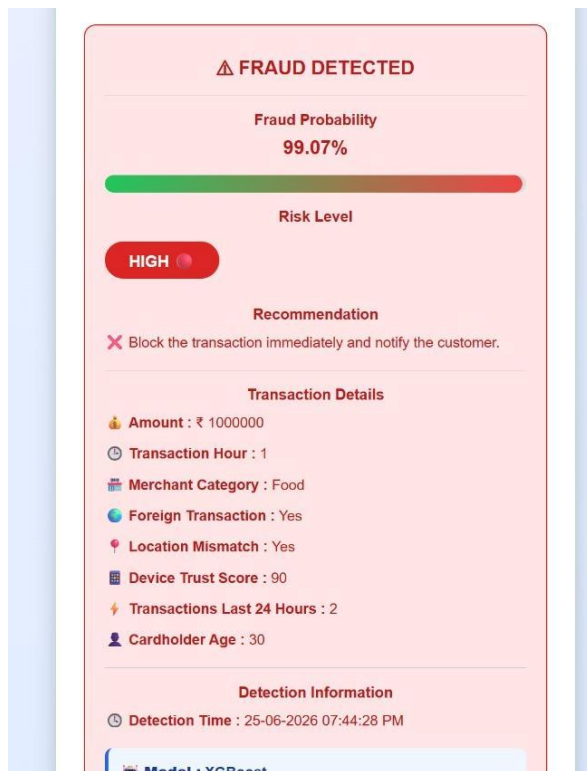
 Cardholder Age : 25

Detection Information

 Detection Time : 25-06-2026 07:47:19 PM

 Model : XGBoost

Fraudulent Transaction:



Challenges Faced:

- Handling highly imbalanced transaction data.
- Selecting appropriate features for model training.
- Integrating the machine learning model with the Flask application.
- Ensuring consistent preprocessing during prediction.

Conclusion:

The Credit Card Fraud Detection System was developed to identify fraudulent transactions using machine learning techniques. The project combines data preprocessing, SMOTE, XGBoost, and Flask deployment to provide an end-to-end fraud detection solution.

The developed application successfully demonstrates how machine learning can be applied to solve real-world problems in the financial domain and provide quick fraud detection support.

Future Scope

- Integration with real-time banking systems.
- Development of an analytics dashboard.

- Implementation of deep learning-based fraud detection models.
- Continuous model retraining using new transaction data.
- Support for large-scale transaction monitoring.

GitHub Repository:

<https://github.com/Venkat-Rao-Bodapati/capstone-project>

References:

1. Kaggle Credit Card Fraud Detection Dataset.
2. Scikit-Learn Documentation.
3. XGBoost Documentation.
4. Flask Documentation.
5. Research articles on credit card fraud detection using machine learning.